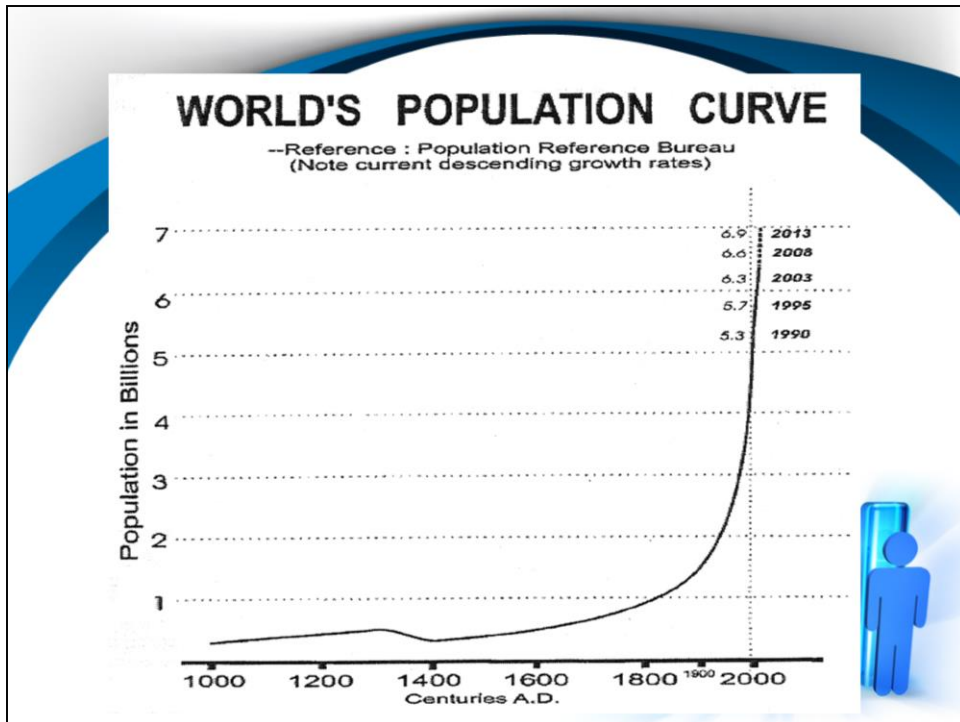


Aging Populations



RRHS APHG 2019-2020



As you know, the world's population is growing. You also know that the growth is not even all around the world. Populations grow (or decline) at different rates depending on a variety of factors – the vast majority of them related to stage of economic and social development (the Demographic Transition Model).

Dependency Ratio

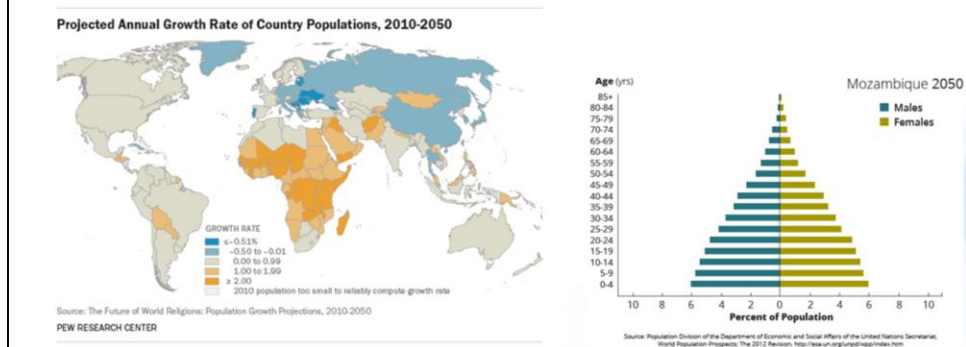
- Ratio of working-age adults to the older (retiree) population and the younger (children) population.
 - **Workers are the “bearing bracket”**
 - Pay into retirement systems
 - Pay taxes to support senior services (healthcare, nursing homes, etc.)
 - Pay taxes to support public education (school tax)
 - Pay tuition for childcare or schooling
 - Pay higher share of taxes toward infrastructure and government operations



As population's growth rates change, there is a shift in the dependency ratio (see notes: Population Composition for more details if you need a review). This particular set of notes is looking specifically at the consequences of “out of balance” populations – those with a high dependency ratio for either the elderly or the youth.

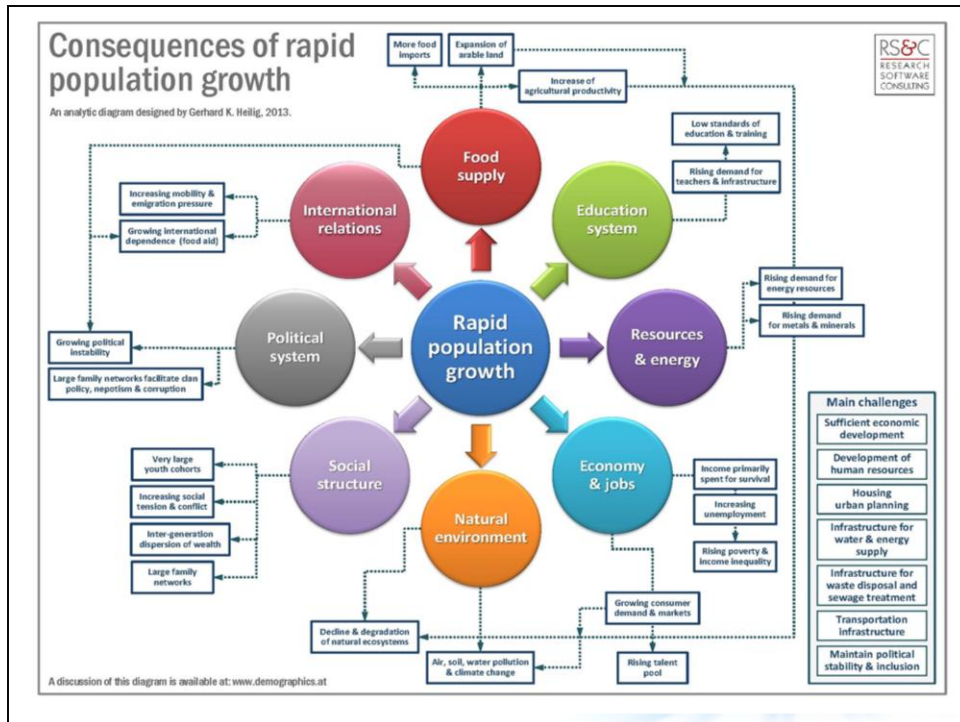
Young Populations

- “bottom heavy” pyramid
 - **Youth outnumber the bearing bracket**
- AKA “Rapid Growth” populations



Countries shaded orange have rapid population growth.

The population pyramid to the right shows a “bottom heavy” population pyramid characteristic of rapid growth countries.



Causes of Rapid Growth

- Decrease in CDR
- Improved health care
- Multiple factors of high fertility existing at once
- High number of migrants, especially rural to urban



Remember, Rapid growth occurs during stages 2 and 3 of the DTM.

To make a population grow quickly, slow down the death rate. This happens through improved healthcare and nutrition. However, at the same time, the total fertility rate is still high because large families are seen as an asset (children provide labor and will care for their parents in their old age).

Take Honduras as an example for multiple factors of high fertility: Honduras combines traditional roles of women (they tend to be homemakers and not hold employment outside the home), male/patriarch dominated society, and Catholicism. All of those factors together contribute to a rapidly growing population.

Rural to urban migration is common around the world as more economic opportunities are available in cities.



Economic Consequences

- Job shortage
- Perpetuation of low-wage jobs
- Income inequality
- Loss of traditional economic practices
- Lack of affordable housing in overcrowded cities



Not all consequences are bad:

A growing youth population means there will be an increasing talent pool of young adults, particularly if education levels continue increasing. Additionally, increasing numbers of people also mean increasing numbers workers for agricultural production (especially in those areas reliant on subsistence farming), which would result in increased agricultural output and an expansion of arable land.

However, the positives are countered by the negatives – increasing numbers of people entering the workforce with limited jobs results in increasing unemployment. When workers are plentiful, wages tend to be lower, which results in a perpetuation of low-wage jobs. Higher demand for housing in urban areas results in higher housing costs and costs of living, which in turn increases the number of people living below the poverty line as the wages do not increase at the same rate. The result is income inequality – the “haves” get more, and the “poor” get poorer.

As more and more people crowd into cities looking for work, there is an increased demand for housing, which drives up the price. City governments cannot keep up with the growth, and “bootstrap” communities may form

(particularly in the most rapidly growing cities of LDCs and NICs) as young adults flood into the cities from the rural areas. These “bootstrap” areas are known as slums – the residents build shelter with whatever materials they have on hand, and they do not have legal claims to the land. These areas also lack infrastructure such as clean water, sewer systems, and electricity.



Social Consequences

- Large family networks
- Educational needs
- Childcare
- Healthcare
- Food insecurity
- Overcrowded cities



One of the biggest benefits of youthful populations is large family networks – these networks provide a built in “safety net” within society as the network is able to help each other in times of individual needs (economic hardship, old age, illness, etc.). Large families are especially beneficial in agricultural-based societies as the children can begin helping with the daily tasks of the farm early in childhood (around 5 or 6, usually), which increases the agricultural productivity of the family.

Larger society challenges, however, include meeting the educational needs of increasing numbers of children each year, especially within the poorest areas of the communities (slums in NICs, rural areas in LDCs, NICs where birthrates are still the highest). There may not be enough well-educated, qualified teachers, money to pay for the buildings/utilities, school supplies, etc. Older girls, especially, are given the roles of tending to the youngest children, so they may not be able to go to school beyond early primary school. And once a girl reaches puberty, if the school does not have adequate sanitation infrastructure, she may not be able to attend classes due to menstruation (which is still considered a taboo topic in most societies, but in some cultures the girl is banned from even being inside the family home and must stay in a “menstruation hut” elsewhere on the property during that time). This natural, cyclic bodily function that is necessary for future reproduction makes it very

difficult for girls in some cultures to continue their education.

For those families not involved in agriculture, there comes the cost of childcare – does one parent (typically the woman) take on the role of a stay-at-home-parent? Do the grandparents mind the children while both parents work? What about paying tuition for the children to get an education? If money is limited, which kid gets to go to school? (The ones most likely to provide for the parents later).

Healthcare is not available for all – whether the distance is too great, the ratio of doctors/midwives to the population is too high, or the cost is too great. Limited access to healthcare increases the rate of infant and maternal mortality, and the child death rate (children dying before age 5, usually due to treatable illnesses such as strep or water-borne disease). As you know, if large numbers of children are dying, the birth rates are higher.

Food insecurity is also a challenge of rapid growth populations as the growth of the population may exceed the population's ability to grow food. This puts an increased strain on the land which may result in environmental issues, as well as the government, which may experience a growing dependence on international food aid. The image above shows Sudanese citizens looking over a delivery of food aid (grains) from the United States.

Political Consequences

- Infrastructure pressures
- Changing voting patterns
- Political and social unrest



In the vast majority of countries, infrastructure projects are paid for through taxes (money paid by citizens to the government, typically based on income).

The government, in turn, is responsible for meeting the needs of the people, including through infrastructure- roads/transportation networks, telecommunications (TV, radio, phones, internet), sanitation (sewage and trash disposal). If a country's population has a high rate of undernourishment (not enough food), the government may be increasingly dependent on food aid from the international community (such as United Nations and World Health Organization).

Changing voting patterns are also of concern in rapid growth countries. Large family networks create issues such as nepotism (the practice among those with power or influence of favoring relatives or friends, especially by giving them jobs), which can also lead to corruption within the government as families with a lot of influence in government may also attempt to hold power and wealth among themselves.

As the population grows rapidly, there is increased mobility and pressure to migrate, both within the country (rural to urban) and internationally. As the

people redistribute themselves, voting patterns change to reflect the interests of the new arrivals in the area.

Finally, political and social movements often start with the young adults – these are the older students and those who see the status quo (the common practices) and speak out against them if they see them as unfair or detrimental to society. The Arab Spring (social revolution in the Middle East and North Africa) that led to the removal of leaders from office started on social media among the youth.

The good news, for governments, is a large number of young people means they have a ready supply of people to serve in their military (which is both good and bad, depending on the circumstances). In many countries, military service is required of all males for a period of time (this was true in Germany and many European countries in the 1800s, and again during the big wars throughout history – World War I, World War II, etc.).

The remaining LDC regions of the world (i.e., the countries of Sub-Saharan Africa), have many tribal disputes and continuous political unrest that stifle economic growth (which needs peace and stability to flourish). This political unrest has many causes, and stunted economic growth and the legacy of European colonialism are contributing factors (more on that in our political unit).



Environmental Consequences

- Land stress
- Deforestation
- Pollution and contamination of soil, air, and water



Rapidly growing populations need food – and to get food they need to increase their arable land. This is done by clearing forests (deforestation), building terraces into mountains (which can lead to erosion and soil loss), and “over-farming” the land by using unsustainable farming practices which cause the soil to lose its fertility. Infertile soil must be replenished with fertilizer (organic such as manure or with chemicals), which can run off into the local water supply and pollute it (too much fertilizer runoff into the local water supply can cause eutrophication, which causes a dense growth of plant life and death of fish and other animal life from lack of oxygen).

The rapidly growing consumer demand in urban areas of LDCs and in NICs can lead to increased factory production, which creates industrial pollution of the air (soot/smog) and water (industrial chemicals dumped into the water). Resource and energy needs also rapidly increase – the mining of raw materials and burning of fossil fuels for energy create environmental damage and pollution of their own. While the long-term impacts of these human activities is unknown, among the scientific community it is generally well-accepted that climate is changing (climate is the long-term averages of weather). The impact of this change disproportionately affects the most marginalized populations – the poor. As climate changes, crop yields are affected. If land once productive becomes damaged, the wealthy have the

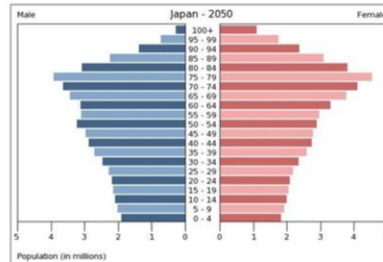
means to move elsewhere while the poor do not.

Aging Populations

- “top-heavy” pyramids
 - **Retirees outnumber the bearing bracket**
- AKA “Declining populations”



3. Population Structures



On the other end of the spectrum of population growth are the “Aging populations”, AKA, declining populations. In these countries the birth rate has fallen below the death rate, and the elderly make up a greater proportion of the population.

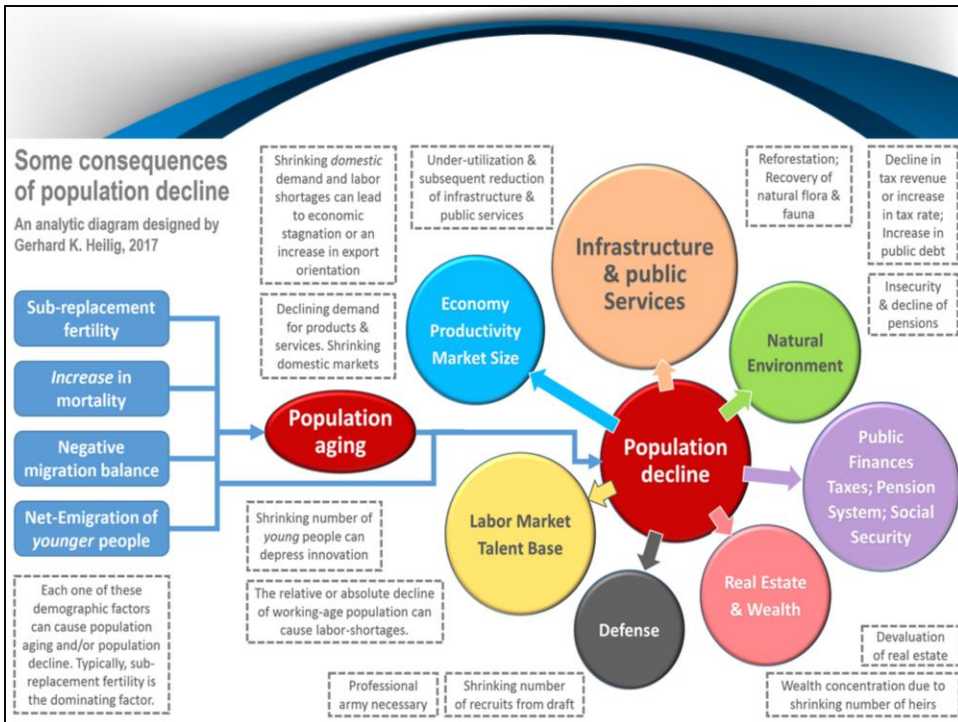
Sometimes the aging population can be caused by younger people leaving the society – this is typical of rural-to-urban migration as the young are more mobile and often leave to look for jobs in the city, and the remaining rural populations have a higher share of older people when compared to urban areas.

Causes of Aging Populations

- Birth rates fall below replacement rate ($TFR < 2.1$)
- Increased life expectancy
- Negative migration balance – young people moving out



The biggest contributing factor to Aging populations is falling fertility rates and increased life expectancy.





One of the biggest challenges of aging populations is that there are not enough young people entering into the workforce to replace the people retiring. The talent pool decreases, which may stifle innovation (if jobs are readily available, fewer people may take the risks of creating their own business or inventing new products as there is little motivation to do so if they can command a high salary from existing jobs).

Fewer workers results in a decline in tax revenues and the “bearing bracket” must pay a higher proportion toward retirement/pension systems to supply an income for the retirees. If there are not enough workers paying into the system, retirement pay for the older people may be cut or even stop – this results in an increasing rate of poverty among the elderly, who by nature have increasing healthcare expenses they may not be able to afford.

Fewer people also results in an over-supply of housing, which drives the prices down (real estate devaluation). Additionally, infrastructure may be underutilized (too few children for a school, too few people using public transportation, etc.) that the cost to maintain that infrastructure may be unsustainable. If there are fewer people, even the local businesses may suffer – grocery stores, for example, need to have a certain number of people in the community in order

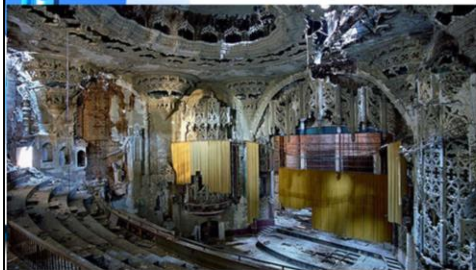
to make it profitable to ship in food for sale. If there aren't enough people buying, the food may go bad before it can be sold.

There is also a consequence for inheritance – with fewer numbers of heirs, the wealth of a nation may become more and more concentrated within a few families.



Social Consequences

- Changing family structure
- Healthcare needs (and costs)
- Retirement systems
- Education



Fewer children results in smaller family networks. Smaller family networks result in fewer people for each individual to “lean on” in times of personal hardship. Children of only children have no aunts, uncles, or cousins. The financial care of the parents (especially if pension payments are limited or healthcare costs are high) fall on one adult child rather than being spread among several children. Suicide rates among the elderly are increasing in some places, and especially at risk are the parents who survive their adult children as there is no one left to care for them after their (adult) child’s death.

Services, such as public schools and hospitals, may be so underutilized that it becomes unsustainable to continue to support them, and they must be closed and consolidated with other facilities – this means that some people will have to travel further to attain these services, which often affects those least able to do so (the poor and the elderly). Pictured above are the urban ruins of an old opera house (left) and a public school (right).

Political Consequences

- Changing voting patterns
- Military service



Politically, governments of aging population face the challenge of both declining tax revenue and increasing demand for expensive services such as healthcare for the elderly. Welfare systems (providing financial, medical, and food assistance for those with incomes below poverty levels) can become overwhelmed.

Voting patterns may also change to reflect the needs of the elderly, who (as expected) will vote for things that suit their needs and desires. In the United States, communities with a higher percentage of older people tend to vote more conservatively – especially rural areas compared to their younger urban counterparts. The map above is an election map by county – It is easy to pick out the location of the U.S.'s large cities as they clearly show as “blue” (Democrat) votes on the map, while the red counties are typically more rural counties (with a few exceptions, such as Lubbock and Amarillo, Tx, which are good sized metro areas that are more conservative).

With declining numbers of youth also comes a reduction in the number of people available to serve in the military. Governments must increase the pay and benefits of military service in order to entice more young people to enlist. The image above is a military recruiting advertisement – appealing to young

adults desire to respected members of the community and a generous cash bonus.



Environmental Consequences

- Reforestation, return to natural state of abandoned areas
- Urban ruins



In another case of “not all consequences are bad” – one impact of aging populations is good for the environment as the land may be left alone and allowed to repair itself – forests regrow, and the flora (plants) and fauna (animal life) is allowed to flourish.

However, there are unique environmental challenges in countries with aging populations as abandoned factory sites and urban ruins (abandoned property in cities) have environmental challenges of their own as they must be cleaned up before they can be redeveloped (more on that in our industrial unit). For example, gas stations usually have buried tanks used to store the gas – if the gas station closes, that tank can fall into disrepair, and any remaining gas in it can leach into the soil, which can make people and animals sick.